BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, BANNU

INTERMEDIATE (12TH CLASS) EXAMINATION, 2022

CHEMISTRY

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

SECTION – A (Version-B)

Time Allowed: 20 Min | **Maximum Marks:** 18

Note: Q.No.1: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct fill that circle in front of the question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero mark in that question.

Q. Which one of the following alkyl halide is least reactive towards nucleophilic substitution reaction?

(A) C_2H_5Br
(B) C_2H_5I
(C) C_2H_5F
(D) C_2H_5Cl

Q. The atomic and molecular masses of different substances can be determined by:

(A) IR spectroscopy
(B) UV/V is spectroscopy
(C) NMR
(D) Mass spectroscopy

Q. The final product of oxidative cleavage of ethylene glycol with periodic acid is;

(A) Formaldehyde
(B) Acetaldehyde
(C) Formic acid
(D) Acetic acid

Q. Schiff base (Imine) can be obtained by reaction of _____ amine with ketone.

(A) Primary
(B) Secondary
(C) Tertiary
(D) Quaternary

Q. The strongest acid among the following carboxylic acid is;

(A) Acetic acid
(B) Formic acid
(C) Propionic acid
(D) Chloroacetic acid

Q. Which one of the following metal sulphate is water insoluble?

(A) $BaSO_4$ (B) Na_2SO_4

(C) $ZnSO_4$ (D) K_2SO_4

Q. Which one of the following is an example of addition polymerization?

(A) Polyvinyl chloride (PVC) (B) Polyesters

(C) Polyamides (D) Nylon 6,6

Q. Which one of the following is not a secondary pollutant?

(A) CO_2 (B) O_3

(C) H_2CO_3 (D) H_2SO_4

Q. Sigma to sigma star transition occurs is;

(A) Alkane (B) Ethene

(C) Ethyl alcohol (D) Methyl chloride

Q. Which one of the following is the biggest source of organic compounds?

(A) Ammonical liquor (B) Coal gas

(C) Coal tar (D) Natural gas

Q. Which of the following chloride have the highest melting point?

(A) NaCl (B) $AlCl_3$

(C) $SiCl_4$ (D) PCl_3

Q. Which of the following Tetrachloride don't hydrolyzed?

(A) $PbCl_4$ (B) $SiCl_4$

(C) CCl_4 (D) None of these

Q. Bronze is an alloy of;

(A) Cu + Zn (B) Fe + Cu

(C) Cu + Sn (D) Mn + Ni

Q. Photochemical smog is primarily caused by:

(A) NO_2 (B) O_3

(C) N_2O (D) CO

Q. The first organic compound prepared in Laboratory was;

(A) Peat (B) Sugar

(C) Urea (D) Propanol

Q. Lassaign's test is used for the identification of _____ in organic compounds.

(A) Carbon
(B) Nitrogen
(C) Hydrogen
(D) Oxygen

Q. The oxidation state of Carbon (C) in Methan (CH_4) is;

(A) +4
(B) -4
(C) -2
(D) +2

Q. The group that deactivate benzene ring and is o/p directing is;

(A) NH_2
(B) -OH
(C) -Cl
(D) $-OCH_3$

SECTION – B

Maximum Marks: 40

Note: Q.No 2. Attempt any TEN parts of the following. All parts carry equal marks.

i. Give reason(s) that why carbonates of group IA elements are more thermally stable than those of group IIA.

ii. Write the chemical reaction of group IIA elements with oxygen and nitrogen.

iii. What is Markownikov's rule? Apply it to the Hydrogenation of propene.

iv. Zn^{+2} salts are white in color while Cu^{+2} salts are coloured why?

v. Differentiate between partial and total synthesis with an example.

vi. Prepare alkene from dehydration of alcohols and dehydrohalogenation of alkyl halides.

vii. Write a chemical reaction of ethene and propene with water.

viii. Write general formula of Grignard reagent. How can you prepare Ethyl Magnesium Bromide?

ix. What is meant by primary, secondary, tertiary alcohols? Give at least one example of each case.

x. How the aldehydes can be differentiated from ketones by means of Fehling and Tollen's reagent test?

xi. Write an equation for Hydrolysis of acetyl chloride and Benzoyl chloride with methanol.

xii. Why Zinc is important for human beings?

xiii. Write the mechanism of SN^2 reaction.

SECTION – C

Maximum Marks: 27

Note: Note: Attempt any THREE of the following. All questions carry equal marks

Q3. Q.No:3.

a) What is meant by an Inert pair effect? What happens, when Tetrachloride of Group IVA reacts with water?

b) Complete the following chemical reactions: i. $[Fe(H_2O)_6]^{+2} + 2NH_3 \rightarrow ?$ ii. $[Fe(H_2O)_6]^{+3} + H_2O \rightleftharpoons$

Q4. Q.No:4.

a) How alkanes are prepared by Clemmenson and Wolf-Kishner reduction?

b) Write notes on the followings. (i) Chemistry of stratosphere (ii) Acid rain

Q5. Q.No:5.

a) Write the systematic names of the following organic compounds. (i) benzene ring bearing NH_2 (aniline) (ii) benzene ring bearing COOH (benzoic acid) (iii) benzene ring bearing CH_3 (toluene) (iv) benzene ring bearing OH and CH_3 (cresol) (v) cyclopentane ring bearing CH_3 (methylcyclopentane)

b) Draw the structural formula of the following organic compounds. (i) 2,3-dimethylpentane (ii) 3-cyclopropyl-2-methylbutane (iii) 1,2-dimethylcyclohexane (iv) Penta-1,3-diene

Q6. Q.No:6.

a) Define mass spectrometry. Explain the construction and working of mass spectrometer.

b) How bromine water test in alkene differ from bromine water test for phenol?

GROUP – II

SECTION-A (VERSION A)

Time Allowed: 20 minutes | **Maximum Marks:** 18

Note: Q.No.1. You have four choices for each objective type question as A, B, C and D. The choice which you think is correct fill that circle in front of the question number in provided bubble sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question.

Q. The atomic size from Na to Cl in the periodic table:

(A) Increase
(B) Decrease
(C) No change
(D) Both A & B

Q. The number of moles of hydrogen gas produced by 1 mole of G-IA elements upon reaction with water:

(A) 0.5 mole
(B) 1 mole
(C) 1.5 mole
(D) 2 mole

Q. There is _____ hybridization in tetrahedrally shaped complex ions:

(A) sp^3
(B) sp^2
(C) sp
(D) dsp^2

Q. The electronic configuration of Cr is:

(A) [Ar], $3d^5$, $4s^1$
(B) [Ar], $3d^5$, $4s^2$
(C) [Ar], $3d^5$, $4s^1$
(D) [Ar], $3d^7$; $4s^0$

Q. –CO– is the functional group of:

(A) Hyroxyl
(B) Ether
(C) Ketone
(D) Alcohol

Q. Each adjacent organic compound of each family is differ due to:

(A) Methyl group
(B) Methane group
(C) Methylene group
(D) None of these

Q. Durene is _____ methylated benzene:

(A) Mono
(B) Di
(C) Tri
(D) Tetra

Q. Which of the following is acidic in nature?

(A) Terminal Alkene
(B) Terminal Alkynes
(C) Non-terminal Alkene
(D) Non-terminal Alkynes

Q. How many isomers are possible for C_2H_6O formula:

(A) 2
(B) 4
(C) 8
(D) 5

Q. Alcohol, phenol and ether are considering as derivatives of the:

(A) CH_4
(B) Oxygen
(C) Hydrogen
(D) Water

Q. An electron pair donor is classified as:

(A) Lewis acid
(B) Lewis base
(C) Bronsted acid
(D) Bronsted base

Q. Oxidation of secondary alcohol gives:

(A) Organic acid
(B) Ether
(C) Aldehyde
(D) Ketone

Q. The number of R groups in tertiary-amines is/are:

(A) 0
(B) 1
(C) 2
(D) 3

Q. Primary, secondary and tertiary-alcohols can be differentiated by using:

(A) Silver mirror test
(B) $KMNO_4$ test
(C) Bromine test
(D) Lucas test

Q. Which of the following is most dangerous for ozone layer?

(A) CFCs
(B) CO
(C) CO_2
(D) NO

Q. Visible radiations ranges from:

(A) (200-800) nm
(B) (400-800) nm
(C) (100-400) nm
(D) All of these

Q. Sucrose is made up of:

(A) Glucose + Fructose
(B) Glucose + Glucose
(C) Fructose + Glucose
(D) All of these

Q. In combustion analysis CO_2 gas is absorbed by:

(A) $Mg(ClO_4)_2$ solution
(B) KOH solution
(C) CuO solution
(D) $CuSO_4$ solution

SECTION-B

Maximum Marks: 40

Note: Q.No.2. Attempt any 10 parts of the following. All parts carry equal marks.

i. The combustion analysis of 0.003 g of Aspirin produced 0.0066g of CO_2 and 0.0012g H_2O. Calculate the empirical formula of aspirin.

ii. Draw the open and cyclic structures of glucose.

iii. Write at least four goals of green chemistry?

iv. How aldehydes and ketones can be prepared from alkenes and alkynes?

v. What are the physical properties of phenol?

vi. How Grignard's reagents react with (a) Esters (b) carbon dioxide?

vii. Why aliphatic amines are stronger bases than NH_3?

viii. Phenol shows acidic characteristics, why?

ix. Describe the test for detection of halogens in a given organic compound.

x. Define Markownikov's rule. Apply this rule to addition of HBr to 1-butene and 2-butene.

xi. What is geometrical isomerism?

xii. Why Cu^{+} ion compounds are diamagnetic while Cu^{2+} ions compounds are paramagnetic?

xiii. How alkali metals react with oxygen?

SECTION-C

Maximum Marks: 27

Note: Note: Attempt any (THREE) questions of the following. All questions carry equal marks.

Q03. Q.NO.03.

a) What is electrophilic aromatic substitution reactions, discuss with mechanism?

b) How coal can be converted into petroleum?

Q04. Q.NO.04.

What are nucleophilic substitution reactions, also discuss its types in detail with mechanism.

Q05. Q.NO.05.

a) What is green house effect? How it is causing global warming?

b) What is the biological role of iron?

Q06. Q.NO.06.

a) Draw structure of following compounds? (i) 2,2-dimethylpropane (ii) But-2-ene (iii) Catechol (iv) Ethanedioic acid (v) 2-methoxybutane

b) Write the names of following compounds according to IUPAC system: (i) benzene ring bearing OH and CH_3 (ii) $(CH_3)_3CC(CH_3)_3$ (iii) HCHO (iv) $CH_2(OH)$-$CH_2(OH)$